Transport Layer Services & Multiplexing Principles

Coverage: Process-to-Process Communication, Port Addressing, Connectionless vs Connection-Oriented Demultiplexing (2-Tuple vs 4-Tuple)

Prepared by: Prof. Anju Markose | Assistant Professor, Dept. of Computer Science & Engineering | VJCET

1. Transport Responsibilities

- Process-to-process delivery using 16-bit Port numbers (0 to 65535).
- Multiplexing & Demultiplexing: UDP uses 2-tuple (Dst IP, Dst Port); TCP uses 4-tuple (Src IP, Src Port, Dst IP, Dst Port).

■ KTU University High-Yield Exam Questions

- Q: Explain how multiplexing and demultiplexing work in UDP and TCP. (8 Marks)